



# अश्वा RACING

DECEMBER 2016 NEWSLETTER





Dear Sponsors, Supporters and Fans,

We wish you all a Merry Christmas and Happy New Year.



This year indeed turned out to be the moment Ashwa longed to cherish. A lot to learn, a lot to venture and a lot to celebrate for.

Finishing 2nd in the design event, 2nd in the Project Management Presentation, at Formula Hybrid New Hampshire left us all with a smile. After bagging the 4th position overall at Formula Hybrid, the combustion team was looking to better the result at Czech Republic. With about 40 teams participating, it was a competitive event. 2nd position in the cost event was our first podium finish and things looked good at that point. Overall we stood 21st. The team is now gearing up for the 2017 season. Valuable lessons learnt will be kept in mind and the momentum in the static events will be maintained.

All of this could be achieved only because of the faith the sponsors have kept in us and the support they have provided us with. A warm thank you to all the well wishers.





# HYBRID UPDATE



Usually when a race-car runs, the roar of the engine is what fills everyone's hearts, both young and old, with exuberance. On the contrary, with our latest Hybrid prototype, the RZ-X7H, the team members were ecstatic when it ran for the very first time and didn't make a sound at all!

If you're wondering why that's good news, the hybrid prototype was run in pure electric mode and its wheels are driven by a 30 kW strong Agni 95R motor. Which means that even the tiniest bit of noise would imply that something is not working the way it's supposed to. The sound of silence is exactly what you want to feel when your Hybrid race-car is in action! And yes, the motor is that silent!

(Note: The above mentioned statements are true for pure electric mode only. The engine still roars like a hungry beast when it's run in Hybrid mode. :))

After nearly 7 months of designing, manufacturing and assembly of the car, our much awaited testing season has finally begun, with preparations going hand-in-hand for the static events as well. Our heartfelt gratitude to everyone of our supporters and well-wishers out there, without whom none of this would've been possible.

Formula Hybrid 2017, here we come!



# SPONSOR FOCUS



Schneider Electric SE is a French multinational corporation that specialises in energy management and automation solutions, spanning hardware, software, and services. Since 2005, the company has authored an annual Planet & Society Barometer, whereby it publicly reports its sustainability metrics in a methodical and transparent way.

Also Schneider Electric recently won:

- 1) Corporate Knights 2016, ranked 12th most sustainable company of the "Global 100 Most Sustainable Corporations in the World"
- 2) Ethisphere's 2016 "One of the World's Most Ethical Companies" for the sixth consecutive year
- 3) Ranked most sustainable company of the CAC 40 for the second consecutive year by Enjeux Les Echos and Institut RSE

We are glad to announce Schnieder Electric being our Platinum Sponsor for the 2016-2017 Race season.





# SPONSOR FOCUS

Dürr AG is a global mechanical and plant engineering firm.,Dürr's activities in India began when the company built a galvanizing line for Tata at Jamshedpur. Today - almost fifty years later - nearly all Dürr business units are active and offer high quality balancing and diagnostic machines, application, paint and final assembly systems as well as environmental systems, industrial cleaning and automation solutions and consulting activities. We thank DÜRR India for their support as gold sponsor for the 2016-2017 combustion race season.



Infineon Technologies AG is a German semiconductor manufacturer founded on 1 April 1999 that make life easier, safer and greener. With a global presence, Infineon offers semiconductors and systems for automotive, industrial, and multi market sectors, as well as chip card and security products. Infineon combines entrepreneurial success with responsible behaviors. Efficient use of energy, environmentally-friendly mobility and security in a connected world - they solve some of the most critical challenges that our society faces while taking a conscientious approach to the use of natural resources. We are pleased to inform you they are part of our 2017 hybrid endeavour as our gold sponsor.

ABB is a Swedish-Swiss multinational corporation headquartered in Zürich, Switzerland, operating mainly in robotics and the power and automation technology areas. They are pioneering technology leaders that works closely with utility, industry, transportation and infrastructure customers to write the future of industrial digitalization. They are ranked 158th in the Forbes Ranking. It's one of the largest engineering companies as well as one of the largest conglomerates in the world. It has operations in around 100 countries.



It's an honour to have them as part of our Ashwa family. We thank ABB for supporting us with their world quality conduits, conduit fittings and lock nuts, pushing us one step closer to our goal.





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**CONTROLWELL**

## SPONSOR FOCUS

With over a decade of experience, Controlwell today is reckoned as the leading distributor of quality Cable Protection and Management products in India. Over the years they have added a wide variety of products ranging from conduits, conduit glands, cable glands, grommets and junction boxes.

We are extremely happy to have Controlwell on board and their timely delivery of components has ensured us to meet our deadlines. We hope for a long lasting partnership with this esteemed company.

element14 is a high-service distributor of technology products, services and solutions for electronic system design, maintenance and repair. With a product range of over 6,00,000 including Single Board Computers, Opensource Hardware, Development Kits, Semiconductors, Circuit Protections, Connectors and Transformers, they bring together the latest products, services, and development software, all connected to an innovative online engineering community. Having been associated with element14 for the past couple of years, Ashwa Racing has not only benefited by the usage of it's state of the art components free of cost, but has also had a lot to take away in terms of the quality of service that element14 has provided the team with. We take this opportunity to extend a heartfelt thank you to the entire team of element14 for going out of their way and ensuring we receive all the components ahead of time, helping the team meet it's deadlines without any hiccups. We hope the partnership bolsters in the coming years with you alongside us throughout our journey.

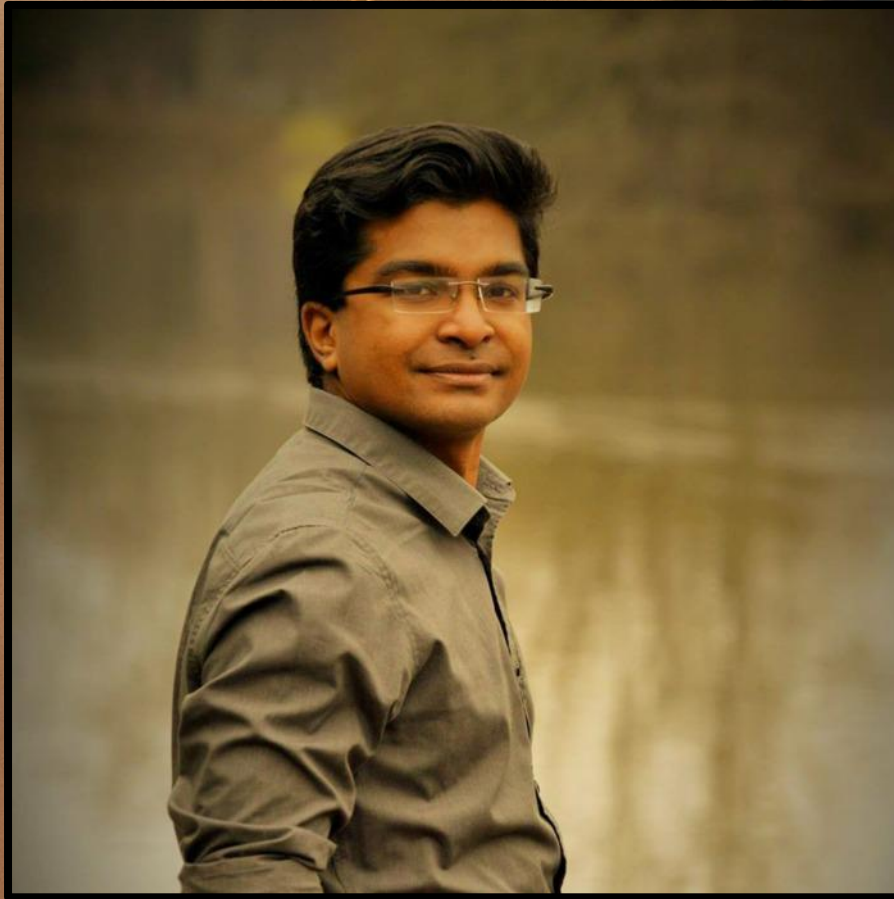
**element14**



Voila! Seat and bodywork of this season are lighter and more beautiful than ever before! We thank ICP India Pvt Ltd for making this possible. With CNC machined patterns for Fiberglass, we have seat and bodywork including air duct for radiator, fabricated with higher accuracy than ever before. ICP India Pvt Ltd is the pioneer in Fiberglass technology in Bangalore. The manufacturing facility located in bommasandra is capable of hand layup, filament winding, vacuum bagging, pultrusion and high precision machining.



# ALUMNI OF THE MONTH



SRIGANESH SRIRAM

Sriganesh Sriram, our alumni of the 2010 batch has been guiding us since he has graduated from RV College of Engineering. Being an Electrical and Electronics Engineering student, Sriganesh was and is an active member of Ashwa Racing. He engaged himself in all the work related to Engine as well as the Electrical subsystem. His intake designs have been considered as a bench mark for all the future batches of Ashwa Racing. Being a student of electrical engineering, his extensive work on the engine, makes him one of the bright members of Ashwa Racing.

Sriganesh worked with Hydraulic Hybrids from 2012 to 2014. He pursued his masters from Georgia Tech. Being a part of the collegiate team called EcoCAR3 funded by the department of Energy and GM, he shared his expertise in converting a Camaro to a hybrid vehicle.





Currently, he's working on fuel cell hybrids at Bosch. He helped the hybrid team extensively with the MATLAB modeling of the hybrid car, one of the key features in the design of the hybrid architecture. He has spent time with the engine and the electrical subsystem of both the hybrid and combustion team, training them with respect to how the design phase should move on and what are the future aspects of these subsystems in the Formula Student arena.

From sharing technical knowledge to contributing towards the funding for the future batch, from coming to the workshop and interacting with us to inviting the whole team to his marriage, we are grateful towards his overall contribution.

We take this opportunity to thank Sriganesh for his continued support and contribution towards the project and helping the project with his knowledge and experience.





# AUDIT - NOVEMBER 2016



R. V. COLLEGE OF ENGINEERING

**ASHWA RACING**

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Ref. No: RVCE/AR/FIN/2016

R. V. COLLEGE OF ENGINEERING **ASHWA RACING**

Accounts for the period from 01-11-2016 to 30-11-2016

| RECIEPTS                                           | AMOUNT                |
|----------------------------------------------------|-----------------------|
| Opening Balance in kotak mahindra Bank RVCE Branch |                       |
| SB A/c 136010046665                                | ₹ 369,601.58          |
| SB A/c 136010058514                                | ₹ 97,376.17           |
| SB A/c 136010060793                                | ₹ 347,083.93          |
| by principal RVCE                                  | ₹ 260,000.00          |
| <b>TOTAL</b>                                       | <b>₹ 1,074,061.68</b> |

| PAYMENTS BY                                        | AMOUNT                |
|----------------------------------------------------|-----------------------|
| Chassis & Suspension                               | ₹ -                   |
| Powertrain & Smp                                   | ₹ -                   |
| Electrical                                         | ₹ -                   |
| Brakes & Logistics                                 | ₹ 5,750.00            |
| Miscellaneous                                      | ₹ 6,009.00            |
| SMS & Wire Charges                                 | ₹ 28.76               |
| Cash In Hand                                       | ₹ 36,575.00           |
| Closing Balance in kotak mahindra Bank RVCE Branch |                       |
| SB A/c 136010046665                                | ₹ 575,807.23          |
| SB A/c 136010058514                                | ₹ 161,524.76          |
| SB A/c 136010060793                                | ₹ 288,366.93          |
| <b>TOTAL</b>                                       | <b>₹ 1,074,061.68</b> |

Checked and Found Correct

*Utkarsh*

Utkarsh Agarwal  
Finance Team  
**ASHWA RACING**  
R.V. College of engineering





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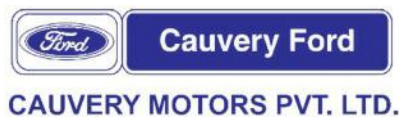
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# Team Directory

## FACULTY

| Name | Designation |
|------|-------------|
|------|-------------|

|                    |                 |
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